

Zhikai Wang

+86-17362254829

| 12410615@mail.sustech.edu.cn

| github.com/Wind645

Research Interest: Efficient Generative Models, Video Generation & Editing, Large Scale Training

EDUCATION

Southern University of Science and Technology (SUSTech)

Shenzhen, China

Bachelor of Science in Computer Science; GPA: 3.94/4.00

Sep. 2024 – Jun. 2028 (Expected)

- **Core Courses:** Data Structures & Algorithms(97), Discrete Mathematics(87), Digital Logic(93), Database Principles(95), Calculus (91), Linear Algebra(95).

SELECTED PUBLICATIONS

FastVMT: Eliminating Redundancy in Video Motion Transfer

(Co-first author), et al. *ICLR* 2026

Z-Cache: Accelerating Diffusion Transformers via Self-Reflection

(Second author), et al. *NeurIPS* 2026

RolloutRemover: Scribble-Guided Object Removal via Latent Rollout Modeling

(Co-first author), et al. *Under review at ICLR 2027*

Flash3DEdit: Escaping Topological Collapse in Accelerated Native 3D Editing via Stage-wise Adaptive Chord Transport

(Co-first author), et al. *Under review at AAAI 2027*

X-Act: Generalizable Action Transfer via Trace-Flow Motion Planner for World Simulation

(Co-first author), et al. *Will submit to CVPR 2027*

A Survey on Cache Methods in Diffusion Models: Toward Efficient Multi-Modal Generation

(Fourth author), et al. *arXiv* 2025 arxiv.org/abs/2510.19755

RESEARCH EXPERIENCE

EPIC Lab, Shanghai Jiao Tong University

Shanghai, China

Research Intern — Advisor: Prof. Linfeng Zhang

Mar. 2025 – Present

Video Diffusion In-Context Ability for Embodied Action Transfer & Editing

Ongoing

- **Research Focus:** Investigating the **in-context capabilities of video diffusion models** for understanding and transforming visual dynamics, with a focus on **embodied action transfer**.
- **Embodied Action Transfer:** Developed **X-Act**, an explicit motion-planning framework that decouples action semantics from source appearance and enables action transfer across different embodiments and scenes.
- **Image Editing:** Developed **RolloutRemover**, a video-diffusion framework that interprets sparse user scribbles as removal intent and performs object removal through structured latent rollout and object-background disentanglement.

FastVMT: Redundancy-Aware Efficient Video Motion Transfer

ICLR 2026

- **Problem Identification:** Analyzed computational redundancy in existing motion transfer methods applied to large video generation models (Wan/DiT).
- **Algorithm Design:** Proposed a **tuning-free** reduction mechanism (**Local Window & Gradient Reuse**) to exploit temporal smoothness.
- **Performance:** Achieved **3.43x inference speedup** and significantly reduced computational overhead while maintaining SOTA visual fidelity.

Efficient Diffusion Deployment (Z-Cache & TaylorSeer)

NeurIPS 2026

- **Research & Deployment:** Investigated efficient inference methods (TaylorSeer, Z-Cache) and engineered their deployment into production-grade diffusion models.
- **Scalability:** Successfully applied caching algorithms to **Stable Diffusion** and **FLUX** pipelines, verifying robustness across architectures.
- **Impact:** Delivered **5.53x speedup** in text-to-image generation tasks with minimal quality degradation (verified via PSNR/SSIM).

TECHNICAL SKILLS

Deep Learning Frameworks: Strong proficiency in PyTorch; familiar with HuggingFace Diffusers
Engineering Tools: WandB, Git, Linux/Bash, Claude Code Agent
Distributed Training: Native PyTorch **FSDP**, **Gradient Checkpointing**, Multi-GPU (A800)
Core Competencies: Diffusion Transformers, Large Multi-Modal model Fine-tuning (LoRA), Video Generation & Cache Inference Optimization

HONORS & AWARDS

Silver Medal, Chinese Physics Olympiad (CPhO)